Srinidhi Shetty

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About Me

I am an Electronics and Communication Engineering undergraduate driven by a deep curiosity to understand how complex systems work from first principles. My interests lie in hardware design, semiconductor physics, solid-state physics, and scientific instrumentation, where I enjoy translating fundamental concepts into practical engineering solutions. Research excites me because it offers the opportunity to investigate challenging problems, continually learn, and contribute to technologies that advance science and engineering.

Education

Dayananda Sagar College of Engineering — Bangalore,India
Bachelor of Engineering in Electronics and Communication CGPA: 7.16/10 — *July 2023 – Sep 2027*

Relevant Coursework: FPGA Prototyping, ASIC Design, VLSI Design, Digital System Design, Microprocessors & Microcontrollers, Signals & Systems, Embedded Systems, CMOS Technology, Semiconductor Physics.

Excellent PU College — Udupi, India
Science: (Physics, Chemistry, Mathematics, Computer Science) Grade: 95.3% — *2021 – 2023*

Experience

Semiconductor Research Trainee — Dec. 2025 – Jan. 2026
Lam Research India – CeNSE, Indian Institute of Science (IISc) — *Bangalore, India*

- Gained hands-on exposure to semiconductor fabrication workflows using SEMulator3D, a process simulation platform widely used in semiconductor technology development.
- Studied semiconductor fabrication process flows, process integration, and device manufacturing using SEMulator3D simulations.
- Strengthened foundational knowledge in semiconductor devices, fabrication technologies, and process engineering.

Hardware Prototyping Lead — May 2025 – Sep. 2025
Samsung Solve for Tomorrow 2025 — *Bangalore, India*

- Led the hardware design and prototyping efforts for an interdisciplinary innovation project selected among the National Top 50 teams in Samsung Solve for Tomorrow 2025.
- Designed and developed the functional hardware prototype, translating conceptual ideas into a working proof of concept for AI-assisted Smart Glasses for the Visually Impaired.
- Contributed to technical planning, system architecture discussions, and prototype validation throughout the competition.

Projects

Verilog-Based AES-128 Hardware Encryption Engine | *Verilog, ModelSim, Vivado* — 2024

- Designed and implemented a synthesizable AES-128 encryption engine in Verilog, incorporating the complete encryption pipeline including SubBytes, ShiftRows, MixColumns, and AddRoundKey transformations.
- Performed RTL simulation and functional verification using ModelSim.
- Synthesized the design for FPGA implementation using Vivado, analyzing timing performance and hardware testing.
- Verified RTL functionality through simulation and evaluated FPGA implementation metrics including timing performance and resource utilization.

Multi Clock Domain Controller IC | *System Verilog, Quartus Prime, EDA Playground* — 2025

- Designed a Multi Clock Domain Controller (MCDC) to enable reliable data transfer between asynchronous clock domains.
- Implemented Clock Domain Crossing (CDC) techniques, including two-stage flip-flop to mitigate metastability.
- Developed robust handshaking mechanisms for safe inter-domain communication and data integrity.
- Verified functional correctness and timing behaviour through simulation, and running different testcases via the testbench codes to check and validate the functionality.

Smart Electric Vehicle Energy Network (SEVEN) | *Peer-to-Peer EV Energy Sharing Platform* — 2024

- Proposed a peer-to-peer electric vehicle energy sharing platform enabling energy exchange between EVs to improve energy accessibility and reduce range anxiety.

- Designed a system architecture incorporating battery health monitoring, state-of-charge estimation, user authentication, and intelligent charging station discovery.
- Researched EV charging infrastructure, V2V/V2G technologies, and sustainable energy management strategies to support future mobility ecosystems.

Drishti – AI Smart Glasses for the Visually Impaired | *System Verilog, Quartus Prime, EDA Playground* 2024

- Conceptualized and contributed to the design of an AI-assisted smart glasses solution aimed at enhancing environmental awareness for visually impaired individuals.
- Led the hardware prototyping efforts, participating in component selection, system integration, and prototype development to validate the core concept.
- Collaborated on product ideation, hardware architecture, and feasibility analysis, resulting in a functional Minimum Viable Prototype (MVP).
- Selected among the National Top 50 teams in Samsung Solve for Tomorrow 2025 for innovation and societal impact.

Technical Skills

Hardware Description Languages: Verilog HDL, System Verilog.
Programming Languages: Embedded C, C/C++, Python.
EDA & Simulation Tools: ModelSim, Vivado, Quartus Prime, Cadence Virtuoso, EDA Playground, Matlab.
Operating System & Tools: VS Code, Linux, PyCharm, Tcl Scripting.
Domains: FPGA Prototyping, RTL Design & Verification, VLSI, Embedded Systems, ASIC Design, CMOS Technology, Physical Design, Hardware Prototyping, Product Design.

Achievements & Awards

National Top 50 (Team Drishti) – Samsung Solve for Tomorrow 2025, hosted by Samsung India under the theme *Environment and Sustainability.*

1st Place (Gold Medal) – Nukkad Natak Competition, IIIT Kharagpur Spring Fest 2026, demonstrating cross-disciplinary excellence.

Special Domain Award – EthicsCare Hardware Hackathon, hosted by DSCE, for innovative hardware design under ethical computing constraints.

Runners-Up – Electronics 1.0 Hardware Hackathon, hosted by IEEE DSCE Chapter and VinyanaTech.

Research Interests

FPGA Applications • Detector Electronics • Sensor Calibration • Digital Hardware Design • Semiconductor Physics

Patents

"Low Cost, Vehicle Independent, Driver Calibrated Brake Drag Detection and Alarm System"
Indian Patent Application – Filed (Under Examination)
Designed an embedded hardware system for real-time brake drag detection using sensor fusion and signal processing techniques for continuous brake drag monitoring and early fault detection.

Certifications

Semiconductor Skills Development Program through SEMulator3D
Lam Research India – Centre for Nano Science and Engineering (CeNSE), Indian Institute of Science (IISc)

CMOS Digital VLSI Design
NPTEL Online Certification, Indian Institute of Technology (IIT) Roorkee

Languages

English (Full Professional Proficiency) • Hindi (Professional Working Proficiency) • Kannada (Native Proficiency)